Assignment Title:

“Write Clean Code and Manage It with Git Version Control”

Problem Statement

You are working as a junior developer on a small software project.
Your task is to **refactor a messy piece of code** into a clean, readable, and maintainable version — following the **Clean Code Principles** (naming, comments, formatting, modularity, etc.).

Then, you must use **Git** to version-control your progress through multiple commits that clearly describe what you changed and why.

Part 1: Clean Code Task

⊘ Original “Messy” Code

Below is a sample **poorly written code**:

It’s intentionally written in a **bad, unreadable, unstructured** way — with:

- No indentation
- Poor variable names
- Hard-coded values
- No comments
- Mixed logic
- Bad formatting

Students’ task will be to **refactor this into clean, modular, readable code** following clean code principles.

```
#include<stdio.h>
int main(){int a[3][3]={{50,60,70},{80,90,100},{30,40,50}};char
n[3][10]={"Ali","Bob","Cat"};int i,j,s;float avg;
for(i=0;i<3;i++){s=0;for(j=0;j<3;j++){s+=a[i][j];}avg=s/3.0;if(avg>=80)p
rintf("%s Grade A avg=%.2f\n",n[i],avg);else if(avg>=60)printf("%s Grade
B avg=%.2f\n",n[i],avg);else if(avg>=40)printf("%s Grade C
avg=%.2f\n",n[i],avg);else printf("%s Fail avg=%.2f\n",n[i],avg);}int
max=0,pos=0;for(i=0;i<3;i++){s=0;for(j=0;j<3;j++){s+=a[i][j];}if(s>max){
max=s;pos=i;}}printf("Topper: %s with total %d\n",n[pos],max);return 0;}
```

✅ Task

Your task:

1. Refactor this code to make it clean and readable.
2. Use **proper naming conventions**, **indentation**, **modular functions**, and **comments**.
3. Handle invalid data gracefully (e.g., empty input or zero marks).
4. Print the result in a neatly formatted way.
5. Push each step to **Git** with clear commit messages.

□ Part 2: Version Control Task (Git)

You must demonstrate proper use of **Git** throughout your project.

Tasks:

1. **Initialize a Git repository**
2. **Make multiple commits**:

   Example Git Commit Sequence

Commit	Description
git commit -m "Added messy version of student marks program"	Initial unrefactored code
git commit -m "Refactored code structure and indentation"	Fixed indentation
git commit -m "Renamed variables for clarity"	Used meaningful names
git commit -m "Added functions for average and grade calculation"	Modularized code
git commit -m "Added comments and formatted output"	Final clean version

3. **Check version history**
4. **Push your repository to GitHub**

□ Submission Requirements

- A GitHub repository link containing:
  - `Abc.c` (clean code)
  - `README.md` (explaining your steps)
  - `git log` screenshot or output file